CollapseCore Campus Network: Project Overview

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Summary

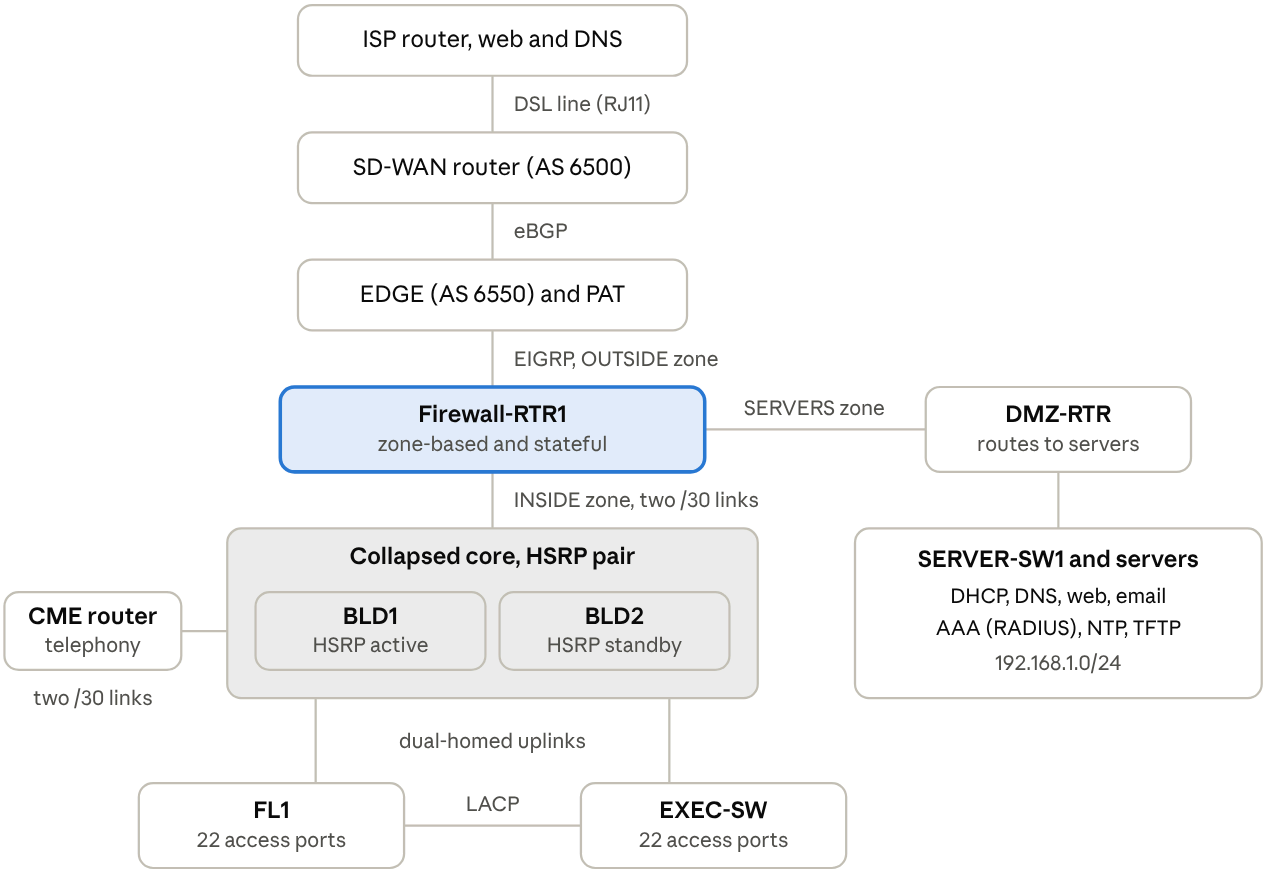
A collapsed-core campus network in Packet Tracer: two Layer 3 core switches provide routing, HSRP and inter-VLAN routing for two access closets, behind a zone-based firewall, a NAT edge router and an SD-WAN path to an ISP. Clients get addresses by DHCP relay, resolve names, browse the intranet and ISP sites, send mail, register IP phones to Cisco CME, and log in to every intranet device over SSH using RADIUS accounts.

- **Scale:** 12 routers and switches, 9 servers, 10 endpoints and 38 links (30 copper, 7 single-mode fiber, 1 RJ11 DSL line).
- **Design approach:** hierarchical, with redundant paths at the core, firewall-to-core and CME layers.
- **Operating model:** services are centralized on a server network, and every intranet device is managed over SSH with RADIUS authentication and a local fallback account.
- **IP scheme:** documented in the accompanying Excel workbook; the key ranges are summarized below.

Topology and hierarchy

The network is built in tiers: an ISP and WAN edge, a zone-based firewall, a collapsed core and an access layer, with the server network hanging off the firewall as its own zone.

Every path between users, servers and the internet crosses the firewall



Laptops, printers and IP phones connect to the access ports; phones use voice VLAN 40.

topology · 12 routers and switches, 9 servers, firewall at the center

A laptop's traffic runs access switch, core, firewall (inside zone), EDGE (where PAT happens), SD-WAN, then the ISP. Servers sit behind the firewall's servers zone, so even internal clients reach them through a policy.

Cabling follows each link's role: single-mode fiber for the core, firewall, CME and WAN-edge links, copper straight-through to servers and endpoints, and an RJ11 DSL line to the ISP cloud.

Addressing and VLAN design

Nine VLANs support the campus: six user-facing subnets from 128.137.232.0/22, each with an HSRP virtual gateway, plus a server network on 192.168.1.0/24 behind the DMZ router. DHCP is centralized: each VLAN's SVI relays to the DHCP server at 192.168.1.2 (`ip`

`helper-address`), and every pool's default gateway is the HSRP virtual IP, not a physical switch address.

VLAN	Purpose	Subnet	HSRP gateway	Notes
30	Client	128.137.232.0/24	.254	BLD1 .1, BLD2 .2
40	VoIP	128.137.233.0/24	.254	DHCP option 150 points phones at the CME
70	Printer	128.137.234.0/27	.30	
80	VIP	128.137.234.32/27	.62	
100	Guest	128.137.234.96/27	.126	Internet-only ACL
999	Management	128.137.234.64/27	.94	Static addresses on the switches (.65 to .68)
200	Servers	192.168.1.0/24	DMZ-RTR .1	Servers .2 to .8; SERVER-SW1 at .254
90 / 600	Native / reserved	none	none	Trunk native VLAN and a reserved "BLK" VLAN

Other address space:

- **Point-to-point links:** /30s in 128.137.234.160 to .183 join the firewall, DMZ router, both cores and the CME.
- **GRE tunnels:** /30s in 128.137.234.188 to .211.
- **Loopbacks:** CME 128.137.234.250 and Firewall 128.137.234.251, both advertised in OSPF.
- **Transit and ISP:** 10.120.100.0/22 (EDGE to SD-WAN), 65.120.100.0/22 (SD-WAN to ISP) and 65.200.100.0/24 (ISP services).

Switching and Layer 3 core

Two Catalyst 3650 switches act as the combined core and distribution layer, and BLD1 is both the spanning-tree root and the HSRP active gateway for every VLAN, so the Layer 2 and Layer 3 paths agree.

- **Inter-VLAN routing:** `ip routing` on both cores, one SVI per VLAN, and HSRP version 2 groups numbered to match each VLAN.

- **Trunks:** 802.1Q trunks carry VLANs 30, 40, 70, 80, 90, 100, 200, 600 and 999 with native VLAN 90. BLD1 and BLD2 are joined by a single-mode fiber trunk.
- **Dual-homed access layer:** FL1 and EXEC-SW each uplink to both cores, so the loss of one core leaves a Layer 2 path.
- **Access ports:** 22 per access switch, with voice VLAN 40, PortFast and BPDU guard. Two ports on each switch form an LACP bundle (Port-channel 1) between FL1 and EXEC-SW.
- **Spanning tree:** PVST, with BLD1 at priority 4096 and BLD2 at 8192 for all VLANs.
- **Voice QoS:** `mls qos` with CoS trust on the access switches.
- **Management:** each switch has a Vlan999 address, with the HSRP virtual IP .94 as its default gateway.

Routing

Three routing protocols are stitched together at two redistribution points, so an intranet host can reach the ISP network and the reverse path works: OSPF inside the campus, EIGRP across the transit link, and BGP to the SD-WAN router.

Segment	Protocol	Devices	Detail
Intranet	OSPF 1, area 0	BLD1, BLD2, Firewall, DMZ-RTR, CME	Both cores originate a default route into OSPF
Intranet to transit	OSPF and EIGRP 1, mutual redistribution	Firewall	EIGRP into OSPF as subnets; OSPF into EIGRP with a set metric
Transit to SD-WAN	EIGRP and BGP, redistribution	EDGE (AS 6550)	eBGP to the SD-WAN router (AS 6500)
Default routing	Static	Firewall, EDGE, SD-WAN, ISP router	Each hop points outward; the ISP router points back to SD-WAN

GRE tunneling. Six GRE tunnels (twelve endpoints) run over the direct /30 links between the firewall, DMZ router, both cores, the CME and EDGE. Each tunnel is sourced from the physical interface with MTU 1476, and the tunnel subnets are kept out of the routing

protocols. All twelve tunnel destinations were checked against the peer's interface address in the saved file.

Path symmetry. The firewall sees two equal-cost routes to every user VLAN, one through each core. With stateful inspection, replies must leave on the interface the request arrived on, so the firewall's link to BLD2 carries an OSPF cost of 10 and BLD1 is the preferred path. This matches BLD1 being the HSRP active gateway.

Redundancy and high availability

The campus layers each have a second path, so losing one core switch or one uplink does not isolate users. The firewall, EDGE and SD-WAN routers are still single devices.

Layer	Mechanism	On failure
Default gateway	HSRP v2 per VLAN; BLD1 priority 105 with preempt, BLD2 at 100	BLD2 takes over the virtual IP
Core uplink	Each core tracks its firewall-facing port; the track decrement is fixed at 10 in this Packet Tracer version, so BLD1 falls to 95, below BLD2	BLD2 becomes active when BLD1 loses its firewall link
Access layer	Each access switch uplinks to both cores; PVST	Traffic moves to the surviving uplink
Firewall to core	A separate /30 to each core, with equal-cost OSPF paths (BLD1 preferred)	The other core carries the traffic
Telephony	CME is dual-homed to both cores and registers phones against a loopback (128.137.234.250) advertised in OSPF	Phones reach the CME over either core link
Default route	Both cores originate a default into OSPF	The CME learns a default from either core

Network services

All shared services run on the 192.168.1.0/24 server network and are reached across the firewall, which keeps them separate from user traffic.

Service	Host	What it does
DHCP	192.168.1.2	Six pools (Client, VoIP, Printer, VIP, Guest, server) served by relay from every SVI; the VoIP pool carries option 150 for the CME
DNS	192.168.1.3	Resolves the intranet resume site and the public ISP site by name
Web	192.168.1.4	Hosts the personal resume page; a separate ISP web server hosts the ISP homepage
Email	192.168.1.5	SMTP and POP3 for the ITSupport.com domain with four mailboxes
AAA	192.168.1.6	RADIUS authentication on UDP 1645 for the intranet devices
NTP	192.168.1.7	Time source with MD5 authentication (key 11)
TFTP	192.168.1.8	File service used by the telephony setup

Telephony. A Cisco 2811 runs CallManager Express (CME), dual-homed to both cores. It has 42 directory numbers (1001 to 1042) with auto-assign, generates the phone configuration files, and sources its phone service from the OSPF-advertised loopback 128.137.234.250. Two Cisco 7960 phones sit on the voice VLAN, get an address and the CME address by DHCP, and register as extensions 1001 and 1002.

Security

Security is layered at four points: how administrators log in, how the access ports behave, what the edge translates, and what traffic the firewall allows between zones.

Device access

- SSH version 2 only on the vty lines, with a shared domain name (ITSupport.com), RSA keys, enable secrets and `service password-encryption` on every intranet device.
- RADIUS AAA on nine devices (SERVER-SW1, DMZ-RTR, Firewall, BLD1, BLD2, FL1, EXEC-SW, EDGE and the CME). Remote logins use `group radius local`, so a local account still works if the server is unreachable.
- The console stays on the local method list, so an AAA or routing problem can never lock out console access.

Layer 2 and edge

- Port security (maximum 5 MAC addresses) on all 22 access ports of FL1 and EXEC-SW, plus PortFast and BPDU guard.
- Trunks use native VLAN 90 and an explicit allowed-VLAN list.
- EDGE performs PAT from the intranet to the transit network, using a standard ACL of the intranet subnets and overloading the WAN-facing interface.
- Guest VLAN 100 has an ACL (`GUEST-IN`) on both cores: DHCP and DNS to 192.168.1.3 are allowed, everything toward 128.137.0.0/16 and 192.168.0.0/16 is denied, and the rest (the internet) is permitted.

Zone-based firewall (Firewall-RTR1)

Three zones map to the physical links: INSIDE (both cores), SERVERS (the DMZ router and server network) and OUTSIDE (EDGE). Traffic between zones is dropped unless a policy permits it. Routing protocols, SSH to the router and GRE terminate on the router itself, so they are not filtered.

Direction	Policy
Inside to Outside	Inspect TCP, UDP and ICMP
Inside to Servers	Inspect DHCP relay, DNS, HTTP and HTTPS, SMTP, POP3, RADIUS, SSH and ping
Servers to Outside	Inspect DNS, web, mail and ping
Outside to Servers	Inspect web, DNS and mail to named servers, plus EDGE's RADIUS requests and ping
Servers to Inside	Pass only the DHCP server's replies (UDP and ICMP from 192.168.1.2)
Outside to Inside	No policy, so everything is dropped

Testing and verification

End-to-end connectivity, services, authentication, telephony and the firewall policy all passed their checks; the failover tests are listed under next steps because they have not been run yet.

Area	Check	Result
DHCP	Clients on the Client and VIP VLANs renew through the relay (`ipconfig /renew`); phones receive VoIP addresses, and VLAN 80 kept working after the CME link was added	Pass
Reachability	Client pings 192.168.1.2 across the firewall	Pass
Web and DNS	Browse the intranet resume page and the ISP homepage by name	Pass
Email	Send and receive between mailboxes	Pass
Time	Devices synchronized to the NTP server	Pass, checked before the firewall policy was applied
AAA	SSH from a laptop to devices using a RADIUS-only account	Pass
Telephony	`show ephone` lists the phones registered with their VoIP VLAN addresses; one phone re-registered after a power cycle	Pass
Routing	Firewall shows three OSPF neighbors in FULL state and a route to every user VLAN through both cores	Pass
Firewall, deny	Server PC and EDGE cannot ping a client	Pass (blocked as designed)
Firewall, allow	`show policy-map type inspect zone-pair sessions` shows inspected sessions and per-class packet counts	Pass

Area	Check	Result
Path symmetry	Web pages loaded slowly until the OSPF cost change, then loaded at normal speed	Pass after fix

Troubleshooting log

Eight problems were found and fixed while building and hardening the network, and each one changed how later changes were made.

Symptom	Root cause	Fix and lesson
Phones got an IP but stayed at "Configuring CM"	The VoIP pool's TFTP server (option 150) pointed at the lab TFTP server, not the CME	Point option 150 at the CME. Phones take their address and their config source from DHCP, so check both.
HSRP would not fail over when the firewall link dropped	HSRP watched only the VLAN interface, and the routed uplink was untracked	Track the firewall-facing port. This Packet Tracer version uses a fixed decrement of 10, so BLD1's priority was lowered to 105 so the decrement drops it below BLD2's 100.
RADIUS configuration commands were rejected on the 3650s	The older `radius-server host` form and a RADIUS source-interface command are not available there	Use the named `radius server` block, and register the real egress addresses (from `show ip route` and `show ip interface brief`) as clients on the server.
A RADIUS login looked like it worked	The same username and password existed locally, so the local fallback answered	Test with an account that exists only on the AAA server.
Zone-based firewall commands were unavailable	The security technology package was disabled on the 2911	Enable `securityk9`, save and reload.

Symptom	Root cause	Fix and lesson
DHCP stopped after that reload	The firewall had lost its OSPF routes to the user VLANs, so replies followed its default route toward EDGE	The inspection session table showed the replies leaving toward the outside zone. `clear ip ospf process` restored the routes.
Web pages loaded slowly through the firewall	Equal-cost paths sent replies through the other core than the request, which stateful inspection treats as a new flow	Raise the OSPF cost on the firewall's link to BLD2 so both directions use BLD1.
Two GRE tunnels showed up/up but pointed at the wrong address	Each destination was the router's own address	A tunnel's destination must be the peer's address on the shared /30. An up/up status alone does not prove a tunnel works, so ping across it.

Skills demonstrated and next steps

The project covers the full path from cabling to a hardened, redundant network, and the troubleshooting log shows the diagnosis skills that sit behind it.

Skills shown

- Hierarchical and collapsed-core design, VLAN and IP planning with summarizable ranges.
- Layer 3 switching, HSRP with uplink tracking, spanning-tree root placement, LACP.
- Multi-protocol routing: OSPF, EIGRP and BGP with controlled redistribution, plus GRE tunneling.
- Centralized services: DHCP relay, DNS, web, email, NTP and a CME voice system.
- Security: SSH, RADIUS AAA with local fallback, port security, PAT, ACLs and a zone-based firewall.
- Method: reading session tables and routing tables to find root causes, and testing with accounts and paths that can't be masked by a fallback.

Next steps

1. **Run and record the failover tests:** shut BLD1′s firewall link during a continuous ping, shut one CME link, and ping across each GRE tunnel while capturing the GRE header in Simulation mode.
2. **Re-check NTP through the firewall:** the zone ACLs do not yet permit UDP 123 to 192.168.1.7, so devices may stop synchronizing.
3. **Tidy the DHCP pools:** several pools still start at .2, which is BLD2′s SVI address, and the Guest pool allows only six clients.
4. **Clean up the CME:** remove the unused SVIs (VLANs 31, 41, 71 and 81) and the static route for the voice subnet, so OSPF carries it over either core.
5. **Harden OSPF:** add passive interfaces on user-facing SVIs and OSPF authentication, one change at a time, checking the firewall′s route table after each.
6. **Publish a server:** add a static NAT on EDGE for the web server to demonstrate a DMZ through the firewall′s outside-to-servers policy.